

THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

ARIN7013: NUMERICAL METHODS IN ARTIFICIAL INTELLIGENCE

May 13, 2024

6:30p.m. – 9:00p.m.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operate satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL 4 questions

Read each question carefully. You may not be credited any score if you put down your conclusion **ONLY**.

Instructions

- Please include the calculations in your solution neatly. This requirement is to ensure that you are not using any mathematics software (which is forbidden).
- You should stop working at 9:00pm.
- Your cheat sheet should be handwritten, double sided A4 paper, otherwise, your cheat sheet will be taken away.

Note: You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully** before you write.

1. (17 points) Let $K(\cdot, \cdot) : \mathbb{R}^4 \times \mathbb{R}^4 \rightarrow \mathbb{R}$ be defined by

$$K(x, x') = \left(1 + \langle \mathbf{x}, \mathbf{x}' \rangle_{\ell^2(\mathbb{R}^4)}\right)^3.$$

Here, $\langle \mathbf{x}, \mathbf{x}' \rangle_{\ell^2(\mathbb{R}^4)}$ denotes the dot product in the Euclidean space $\mathbb{R}^4$. Let $\{(\mathbf{x}_i, y_i), i = 1, \dots, 4\} \subset \mathbb{R}^4 \times \mathbb{R}$ be a set of training data. Here, $\mathbf{x}_1 = [1, 0, -1, 0]'$, $\mathbf{x}_2 = [-1, 0, 0, 1]'$, $\mathbf{x}_3 = [0, 0, 0, 0]'$ and $\mathbf{x}_4 = [0, 1, -1, 0]'$, $y_1 = y_2 = y_4 = 2$ and $y_3 = 0$.

- (a) (2 points) Recall that the Gram matrix $\mathbf{K} \in \mathbb{R}^{4 \times 4}$ is defined by $\mathbf{K}_{i,j} = K(\mathbf{x}_i, \mathbf{x}_j)$ for $i, j = 1, 2, 3, 4$. Calculate the Gram matrix $\mathbf{K}$.
- (b) (5 points) Use the power iteration method to find the largest eigenvalue of the Gram matrix $\mathbf{K}$ by starting at $\mathbf{x}_0 = [1, 0, 0, 0]'$. Show the result of the first three iterations.
- (c) (5 points) Let $\mathbf{y} := [y_1, y_2, y_3, y_4]'$, $\alpha = 0.5$, and $I = \text{diag}(1, 1, 1, 1)$. Can you use Doolittle LU factorization to obtain the LU decomposition of $\mathbf{K} + \alpha I$? If so, then find the LU decomposition. Hence, calculate $(\mathbf{K} + \alpha I)^{-1} \mathbf{y}$ without inverting the matrix.
- (d) (5 points) Let $(\mathcal{H}, \|\cdot\|_{\mathcal{H}})$ be the corresponding RKHS for $K(\cdot, \cdot)$. Kernel Ridge Regress (KRR) solves for

$$\arg \min_{f \in \mathcal{H}} \frac{1}{4} \sum_{i=1}^4 |f(\mathbf{x}_i) - y_i|^2 + \beta \|f\|_{\mathcal{H}}^2$$

with $\beta > 0$ being the penalty parameter. Find the objective function $f : \mathbb{R}^4 \rightarrow \mathbb{R}$ with $\beta = 0.125$.

2. (33 points) Let $\mathbf{x}_1 = [1, 0, -1, 0]'$, $\mathbf{x}_2 = [-1, 0, 0, 1]'$, $\mathbf{x}_3 = [1, 1, 0, 0]'$ and $\mathbf{x}_4 = [0, 1, -1, 0]'$.

(a) (15 points)

- i. (5 points) Find the first principal component of this set of data using the inverse iteration starting with the initial guess of the largest eigenvalue at $\nu = 1$ and the initial eigenvector $\mathbf{v}_0 = [1, 0, 0, 0]'$. Show the result of the first three iterations.
- ii. (5 points) Find the second principal component of this set of data using the inverse iteration starting with the initial guess of the second largest eigenvalue at $\nu = 0.4$ and the initial eigenvector $\mathbf{v}_0 = [1, 0, 0, 0]'$. Show the result of the first three iterations.
- iii. (5 points) Denote the first two principal components as $\mathbf{v}_1$ and $\mathbf{v}_2$. Calculate the mean square error of this set of data onto the first two principal components,

$$\frac{1}{4} \sum_{i=1}^4 \left\| \mathbf{x}_i - \sum_{j=1}^2 \langle \mathbf{x}_i, \mathbf{v}_j \rangle_{\ell^2(\mathbb{R}^4)} \mathbf{v}_j \right\|_2^2.$$

- (b) (18 points) Let $K(x, y) = \langle x, y \rangle_{\ell^2(\mathbb{R}^4)}^2$ for $x, y \in \mathbb{R}^4$ be the kernel function. Let $\mathcal{H}$ be the corresponding RKHS for the kernel $K(\cdot, \cdot)$, and let $\Psi : \mathbb{R}^4 \rightarrow \mathcal{H}$ be the kernel map. We plan to apply kernel PCA to find the principal components from the same data points, i.e., we want to apply PCA to the following set of functions in $\mathcal{H}$,

$$\{\Psi(\mathbf{x}_1), \Psi(\mathbf{x}_2), \Psi(\mathbf{x}_3), \Psi(\mathbf{x}_4)\}.$$

- i. (3 points) Calculate the mean of the functions,

$$E := \frac{1}{4} \sum_{i=1}^4 \Psi(\mathbf{x}_i).$$

- ii. (5 points) Define the Gram matrix $\mathbf{K} \in \mathbb{R}^{4 \times 4}$ by

$$\mathbf{K}_{ij} = \langle \Psi(\mathbf{x}_i), \Psi(\mathbf{x}_j) \rangle_{\mathcal{H}}$$

for $i, j = 1, \dots, 4$. Calculate this matrix.

- iii. (5 points) Use the Rayleigh quotient iteration to estimate the first principal component of this kernel PCA, i.e., estimate the largest eigenvalue of the Gram matrix $\mathbf{K}$, with initial guess $[0.5, 0.5, 0.5, 0.5]'$. Present the first three iterative solutions.
- iv. (5 points) Denote the first principal component as $\mathbf{v}$. Calculate the mean square error of this set of data onto this first principal component,

$$\frac{1}{4} \sum_{i=1}^4 \|\Psi(\mathbf{x}_i) - \langle \Psi(\mathbf{x}_i), \mathbf{v} \rangle_{\mathcal{H}} \mathbf{v}\|_{\mathcal{H}}^2.$$

3. (22 points) Consider a Residual net block,

- INPUT. 100×100 grayscale images: X
 - LAYER 1. Convolutional layer with one 5×5 convolutional filter K_1 . Here, we set the stride length = 1. To capture the boundary of the input image, we use padding by adding 2 rows on the up, 2 rows on the bottom, and 2 columns on the left, 2 columns on the right with zero elements around it, which is denoted by X_1 .
 - LAYER 2. ReLU
 - LAYER 3: Convolutional layer with one 5×5 convolutional filter K_2 . Here, we set the stride length = 1. To capture the boundary of the input image, we use padding by adding 2 rows on the up, 2 rows on the bottom, and 2 columns on the left, 2 columns on the right with zero elements around it, which is denoted by X_2 .
 - OUTPUT LAYER: adding X to the output from LAYER 3.
- (a) (7 points) Let the output be Y , then it can be expressed as a function of the input X . Can you formulate the expression of Y using X ?
- (b) (7 points) Denote θ as the weights involved in this Residual net block. Name and calculate the number of weights in the expression $Y(X; \theta)$.
- (c) (8 points) Suppose you want to complete a learning task on classifying a large number of grayscale images, say 1 million, of dimension 100×100 into two categories: Cat or Dog. In other words, you want to label all these 1 million images by -1 or 1 , with -1 meaning this image is an image of a cat and 1 for the dog. Provided with N number of training data: $\{(x_i, y_i)_{i=1}^N\}$ and $y_i \in \{-1, 1\}$ for $i = 1, \dots, N$. To complete this task, you can sum up all the components of the output $Y(X; \theta)$ to obtain a scalar, denoted as $\tilde{Y}(X; \theta)$.

Now, you want to train your residual net block by

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^N \left(\text{sign}(\tilde{Y}(x_i; \theta)) - y_i \right)^2.$$

Here, $\text{sign}(x) = 1$ if $x > 0$ and $\text{sign}(x) = -1$ if $x \leq 0$.

Can you provide the methodology for finding the optimal parameter θ ?

4. (28 points) Let $f(t, x) : (0, 1] \times [x_0 - b, x_0 + b]$ be a bivariate function with bounded partial derivatives up to order 3. Consider the following initial value problem,

$$\frac{dx}{dt} = f(t, x) \text{ for } t \in (0, 1], \quad x(0) = x_0.$$

For $N \in \mathbb{N}$, we divide the interval $[0, 1]$ by the $N + 1$ mesh points,

$$0 = t_0, t_1 = h, \dots, t_N = Nh = 1,$$

with $h := 1/N$ as the step size.

- (a) (18 points) Consider the following numerical scheme to get the approximation of $x(t_i)$ for $i = 1, \dots, N$ sequentially, by

$$x_i = x_{i-1} + \frac{h}{2} \left(f(t_{i-1}, x_{i-1}) + f(t_i, x_{i-1} + hf(t_{i-1}, x_{i-1})) \right).$$

x_i denotes the numerical solution to approximate $x(t_i)$.

- i. (3 points) Is this numerical scheme explicit or implicit, and why?
- ii. (5 points) The truncation error is defined by

$$T_i := \frac{x(t_i) - x(t_{i-1})}{h} - \frac{1}{2} \left(f(t_{i-1}, x(t_{i-1})) + f(t_i, x(t_{i-1}) + hf(t_{i-1}, x(t_{i-1}))) \right).$$

Can you derive an upper bound for T_i when the mesh size $h \rightarrow 0$?

- iii. (5 points) Let $e_i = x(t_i) - x_i$ be the discrepancy between the exact solution and the numerical solution at t_i . Calculate

$$\begin{aligned} \frac{e_i - e_{i-1}}{h} - \frac{1}{2} \left(f(t_{i-1}, x(t_{i-1})) + f(t_i, x(t_{i-1}) + hf(t_{i-1}, x(t_{i-1}))) \right) \\ + \frac{1}{2} \left(f(t_{i-1}, x_{i-1}) + f(t_i, x_{i-1} + hf(t_{i-1}, x_{i-1})) \right). \end{aligned}$$

- iv. (5 points) Hence, derive an upper bound for

$$\max_{i=0,1,\dots,N} |e_i| \text{ when } h \rightarrow 0.$$

(b) (10 points) Let

$$\frac{dx}{dt} = t - x^2, \quad x(0) = 0.$$

- i. (5 points) Use Forward Euler scheme with mesh size $h = 0.25$ to calculate the numerical solution x_i for $i = 1, 2, 3$.
- ii. (5 points) Use the numerical scheme in 4(a) with mesh size $h = 0.25$ to calculate the numerical solution x_i for $i = 1, 2, 3$.

***** **END OF PAPER** *****